A Rigorous Testing Manual for 16 Classical and Geometric Segmentation Approaches

How to test each method without invalidating it by proxy, oracle leakage, wrong-direction scoring, or metric mismatch

Purpose: This manual is written for an experimental agent. Its job is not to defend or kill an idea quickly. Its job is to test the actual hypothesis that makes the method useful, implement the smallest faithful version of the operator, and measure whether that operator improves Panoptic Quality without unacceptable collateral damage.

Prepared from the current experiment history (exp088-exp110) and the 16-method proposal.

Contents

Part I: Non-negotiable experimental protocol
Part II: Known facts that should constrain, not predetermine, the tests
1. Geodesic / shortest-path completion
2. Elastic rod / Euler-elastica
3. Level sets / active contours
4. CRF / graph-cut refinement
5. Vesselness / ridge filtering
6. Orientation fields
7. Persistent homology / topological persistence
8. Minimum-cost flow / global filament assignment
9. Matched filtering
10. Anisotropic / coherence-enhancing diffusion
11. Distance-transform watershed
12. Medial-axis width-profile analysis
13. Shape derivatives / boundary sensitivity
14. Optimal transport between predictions
15. Test-time energy minimization
16. Combined endpoint geodesic + orientation support

Appendix A: Standard experiment report schema
Appendix B: Decision tree and stopping rules
Appendix C: Checklist before declaring a method invalid

Part I - Non-negotiable experimental protocol

Core rule: A method is not invalidated because a convenient scalar proxy looks weak. A method is invalidated only when a faithful, minimally tuned implementation of its decision rule fails to improve the target metric, or when a necessary precondition is tested with a measurement that is actually equivalent to that precondition.

1. Separate the four layers of every experiment

1. Failure population: defined with ground truth for analysis only (for example, strict 1:1 pairs with IoU 0.40-0.50, or 1:N merge components). This is an oracle-labelled research population, not an inference-time trigger.
2. Observable trigger: information available at inference time that decides whether the operator should run (for example, a width saddle, another model extending beyond the boundary, or a low-cost orientation-consistent path).
3. Repair operator: the actual transformation applied to the prediction (for example, shortest-path extension, graph cut, watershed split).
4. Evaluation: recompute the official matching and PQ globally after the repair. Never infer PQ gain from AUC, Dice, or a local surrogate alone.

2. Always distinguish an oracle ceiling from a deployable method

Oracle experiments are allowed and valuable, but only as upper bounds. They answer whether a repair class could matter if the correct instances and correct direction were known. They do not provide an inference rule.

```
Oracle ceiling  !=  trigger quality  !=  actual deployed PQ gain
```

- For a merge oracle: use truth to split a merged prediction and measure the maximum possible rescues. Label the result ORACLE explicitly.
- For a growth oracle: use truth to identify the missing region, but never call that a practical postprocessor.
- Only after an oracle shows material headroom should you test an observable trigger and a faithful operator.

3. Test the statistic that the algorithm actually uses

Example: A geodesic chooses a minimum-cost path. The mean probability in a cone is not equivalent to the cost of the best path. Therefore a low cone mean cannot invalidate geodesic completion. Likewise, width coefficient of variation cannot invalidate a local saddle detector, and global Hessian selectivity cannot invalidate ridge evidence sampled only along one endpoint corridor.

5. Write the decision rule mathematically before writing the audit.
6. Identify the statistic that is sufficient for that rule. If the method performs an optimization, run that optimization or compute the objective it would minimize.
7. Only use a cheaper proxy if you prove or justify that failure of the proxy necessarily implies failure of the actual method.
8. If the proxy is merely correlated with the true criterion, report the result as partial evidence, never invalidation.

4. Score both feature directions

For every scalar feature s, evaluate both s > t and s < t. If ROC AUC is below 0.5, report the flipped AUC 1-AUC and test the reversed threshold direction. AUC 0.397 is not evidence of no signal; it corresponds to 0.603 ranking information in the opposite direction.

```
AUC_flipped = max(AUC, 1 - AUC)
```

5. Validate the instrument before interpreting it

Whenever possible, construct a ground-truth-only label statistic that should separate the populations if the measurement geometry is correct. Example: for endpoint continuation, measure truth occupancy along the best forward ray. If that label itself does not separate truncated from healthy instances, the endpoint/tangent/cone instrument is invalid and no feature result should be trusted.

6. Use matched controls and avoid prevalence traps

- Healthy controls should be drawn from the same component class, size range, morphology range, and image distribution as the failure population whenever possible.
- When the positive class is rare, ROC AUC can look respectable while precision is useless. Always report precision-recall behavior and the number of healthy instances harmed per true positive repaired.
- Report the positive prevalence. A trigger with 15% precision can be good or terrible depending on the base rate and the severity of false repairs.

- Where feasible, bootstrap images rather than individual instances so confidence intervals respect within-image correlation.

7. Measure the official metric after every operator

The final arbiter is the frozen official scorer. For PQ: rebuild the global matching, then report TP, FP, FN, IoU numerator, denominator, and PQ. Local IoU improvements can change global assignments, so do not approximate the final effect when an exact scorer exists.

```
PQ = sum(IoU over matched pairs) / (TP + 0.5*FP + 0.5*FN)
```

Threshold cliff: An unmatched prediction-truth pair that becomes a TP at IoU 0.51 keeps denominator contribution at 1 but adds about 0.51 to the numerator. This is why crossings matter much more than polishing an already matched 0.55 object to 0.60.

8. Mandatory harm accounting

- Number of targeted positives repaired.
- Number of targeted positives made worse.
- Number of healthy matched objects harmed enough to fall below IoU 0.5.
- Number of new FP/FN pairs created by splitting or growth.
- Change in mean IoU among already matched objects.
- Net PQ, not just rescue count.

9. Parameter-search discipline

Small parameter sweeps are acceptable for deterministic algorithms, but avoid turning the validation set into a hidden training set. Use coarse, physically interpretable grids first; freeze a candidate; then verify on a held-out subset or by grouped cross-validation if the dataset allows it. Record every attempted configuration.

10. Required language in reports

Use	"not tested", "partially tested", "precondition weakened", "oracle ceiling", "operator tested and negative"
Do not use	"invalidated", "dead", "no signal", "cannot work" unless the actual method or an equivalent necessary condition was tested
When AUC < 0.5	Report the reversed direction and flipped AUC
When a proxy fails	State exactly what it rules out and what remains open
When counts change	Explain why the population changed; otherwise treat it as a bug

11. Minimum standard report for every method

```
method: "..."
hypothesis: "..."
failure_population: "..."       # truth-defined, analysis only
control_population: "..."
observable_inputs: [ ... ]        # inference-time only
oracle_ceiling: {...}             # if measured
instrument_validation: {...}
trigger_features: {...}
feature_directions_tested: ["high", "low"]
operator: {...}
parameter_grid: {...}
official_before: {pq, tp, fp, fn, numerator, denominator}
official_after:  {pq, tp, fp, fn, numerator, denominator}
threshold_crossings_up: N
threshold_crossings_down: N
healthy_objects_harmed: N
positive_objects_repaired: N
conclusion_scope: "exactly what this experiment rules out"
status: "untested | partial | negative | positive | inconclusive"
```

Part II - Known facts that should constrain, not predetermine, the tests

The current experiment history contains useful measurements. They should determine where to spend effort, but they must not be used to declare an unimplemented method invalid.

Baseline reproduction	A prior audit reproduced PQ about 0.45586 with TP/FP/FN 860/397/465. Recompute from current artifacts before every new audit rather than trusting cached counts.
Near-threshold structure	The large majority of near-threshold 1:1 failures were shape errors rather than split/merge topology errors. This demotes global graph decomposition as the first move but does not invalidate split/merge methods.
Axial vs lateral deficit	exp108 reportedly found about 59% of missing area axial and 41% lateral. This makes endpoint methods relevant and narrow-band boundary methods incomplete, not useless.
Semantic field	exp106/110 found weak or reversed semantic support around truncated objects. Treat semantic-only continuation as weak evidence, not proof that image structure is absent.
Raw absorption	exp110 found dark_best_ray around AUC 0.611. That is weak as a trigger but enough to keep image-derived endpoint evidence open.
Ridge global proposals	exp103 found poor global selectivity: retaining 80% of filament pixels accepted about 26.2% of the solar disk. Local endpoint ridge evidence remains untested.
Width statistic	rho = w_min / median(w) reportedly reached AUC about 0.712 but low precision. This says width structure may contain signal, while one scalar gate is insufficient.

Data hygiene: The logs have shown small count discrepancies across audits (for example 55 vs 56 near-threshold cases). Every new script must reconstruct the population from source artifacts and assert expected invariants instead of copying counts from prose.

1. Geodesic / shortest-path completion

Current evidence status: Partially screened; semantic-field-only evidence is weak, but the actual shortest-path operator has not been fairly ruled out.

Hypothesis	Truncated 1:1 instances have a physically plausible continuation from a skeleton endpoint through an image-derived cost field. A shortest path can extend the instance more selectively than dilation.
Target population	Strict 1:1 pairs with IoU 0.40-0.50, further restricted to cases whose missing truth lies substantially beyond a skeleton endpoint. Healthy strict 1:1 controls matched by length/area and image.
Inference-time inputs	Predicted instance mask, skeleton endpoint and endpoint tangent, raw image channels, optional semantic/spine probabilities, optional ridge/orientation field. No ground truth at inference.

Instrument validation

- For each endpoint, define a forward search region (e.g. angular cone plus maximum physical length). Validate the endpoint and cone geometry using a truth-only statistic: truth occupancy along candidate paths must be higher for truncated positives than controls.
- Do not use cone mean as the decisive statistic. The algorithm chooses a path; therefore compute path costs.

Faithful implementation

9. Define pixel cost c(x). Start simple: c = a*(1-dark_support) + b*(orientation_mismatch) + c0. Add semantic/ridge terms only in ablations.
10. Construct a pixel or superpixel graph inside the forward ROI. Edges connect 8-neighbours; optionally add turning penalties by augmenting state with direction.
11. Run Dijkstra / A* / fast marching from the endpoint to candidate terminal states at increasing distances. Store minimum integrated cost, maximum gap cost, path length, cumulative support, and curvature.
12. Convert the selected path into mask area using endpoint width (distance transform) or a local width profile; do not score only the centreline.
13. Gate the operator using inference-time path quality, not truth-defined near-threshold status.

Minimal viable experiment

Implement one endpoint, one cost field (raw dark support + turning penalty), one path-length range, and one width reconstruction rule. Compare actual PQ against baseline and against uniform endpoint extension of the same average added area.

Controls / ablations

- Random-direction path with same length/width (tests whether any added area helps).
- Straight tangent extension (tests whether Dijkstra is adding value).
- Uniform dilation matched by added pixel count (tests directional selectivity).
- Healthy endpoint population to quantify false extensions.

Required metrics

- Official PQ delta
- Number of 0.4-0.5 cases crossing 0.5
- Number of healthy matches pushed below 0.5
- Path cost distribution on positives vs controls
- Added-area efficiency: IoU gain per added pixel

Decision rule

Advance / accept: Advance if an inference-time gate has useful precision and the faithful path operator produces positive PQ with stable gains across images. A weak AUC alone is insufficient.

Reject / demote: Reject a specific cost formulation if path selection plus mask reconstruction is net-negative across a reasonable parameter grid. Do not reject all geodesics because semantic support contains zeros.

Common invalid conclusions to avoid

- Using mean cone support instead of minimum path cost.
- Treating P=0 as an impassable pixel unless the implementation actually sets infinite cost.
- Evaluating centreline overlap but not reconstructed mask PQ.
- Letting truth select which endpoint or stopping distance to use in the deployed result.

2. Elastic rod / Euler-elastica continuation

Current evidence status: Untested. It should be viewed as a smoothness-regularized continuation model, not assumed equivalent to geodesics.

Hypothesis	Missing filament continuation is smooth in tangent and curvature. Combining image evidence with bending energy suppresses spurious zig-zag paths through background structure.
Target population	Same axial-truncation population as Method 1, but particularly cases where several image-supported routes compete.
Inference-time inputs	Endpoint position, tangent, optional estimated curvature, image/support cost field.

Instrument validation

- Confirm that truth continuation direction/curvature is predictable from the final observed segment. Measure tangent error and curvature continuity against truth as an oracle diagnostic.

Faithful implementation

14. Parameterize continuation gamma(s) for s in [0,L]. Use energy E = lambda_b * integral kappa(s)^2 ds + lambda_d * integral c(gamma(s)) ds + optional endpoint/length terms.
15. Use a discrete chain of control points or cubic B-spline. Fix the first point and initial tangent from the observed skeleton. Optimize remaining control points with dynamic programming, L-BFGS, or a small beam search over headings.
16. Constrain maximum curvature and search corridor. Reconstruct mask width along the curve from the existing width profile.
17. Sweep only a few interpretable lambda_b values: nearly straight, moderate bending, flexible.

Minimal viable experiment

Compare three operators on the same endpoints: straight extrapolation, shortest path without curvature, and elastica-regularized path. If elastica does not beat both, its complexity is not justified.

Controls / ablations

- Truth-only best smooth curve as an oracle ceiling.
- Randomized tangent perturbations.
- Same image term with lambda_b=0.

Required metrics

- PQ delta
- Crossings
- Mean angular error of continuation where truth is available
- False continuation rate
- Curvature of selected paths versus healthy filament curvature distribution

Decision rule

Advance / accept: Keep if curvature regularization improves PQ or false-extension rate over pure geodesic at comparable rescue count.

Reject / demote: Reject the elastica regularizer if it consistently suppresses true recoveries or does not improve geodesic/straight baselines. This does not reject endpoint completion itself.

Common invalid conclusions to avoid

- Declaring it invalid because semantic geodesics are weak.
- Optimizing only curve shape without rasterizing to the mask width.
- Using ground-truth curvature as an inference input.

3. Level sets / active contours

Current evidence status: Untested as an operator. The axial/lateral audit suggests it targets only part of the error unless initialized with a larger capture region.

Hypothesis	For instances that are laterally too thin or locally misaligned, evolving the boundary under data and curvature forces can recover area more selectively than uniform dilation.
Target population	Strict 1:1 failures with predominantly lateral missing area; optionally a secondary axial group to test whether region-based forces can extend endpoints.
Inference-time inputs	Initial mask, image/semantic likelihoods, edge map, optional orientation map.

Instrument validation

- Classify missing truth pixels into lateral versus axial relative to the skeleton. Verify that the chosen positive population actually has lateral deficit.

Faithful implementation

18. Initialize signed-distance phi from the prediction mask.
19. Use a minimal force: region/semantic outward force + curvature regularization + edge stopping. Example dphi/dt = alpha*F_data*|grad phi| + beta*kappa*|grad phi|.
20. Limit evolution to an ROI around the instance to prevent leakage into distant network structures.
21. Run a small fixed number of iterations or stop on energy convergence. Record area change and boundary displacement.
22. Test separate outward-only and bidirectional versions; the latter can shrink false excess while growing deficits.

Minimal viable experiment

Run on the lateral-deficit subset with one region term and one curvature term. Compare with equal-area morphological dilation/closing.

Controls / ablations

- No curvature term
- No data term
- Uniform dilation of matched area
- Healthy 1:1 masks

Required metrics

- PQ delta
- Boundary F-score or IoU only as diagnostics
- Crossings up/down
- Leakage area into neighbouring structures
- Fraction of healthy masks degraded

Decision rule

Advance / accept: Advance if local contour evolution improves lateral cases and net PQ with substantially less collateral damage than dilation.

Reject / demote: Reject a force formulation if all reasonable stopping times are net-negative. Do not use the 59% axial statistic to reject lateral active contours.

Common invalid conclusions to avoid

- Applying globally to all masks.
- Evaluating only Dice.
- Allowing the contour to see truth-derived stopping time.
- Interpreting non-improvement on axial cases as failure on lateral cases.

4. CRF / graph-cut refinement

Current evidence status: Untested as a repair. Narrow-band graph cuts are demoted for axial errors but remain plausible for lateral boundary error.

Hypothesis	A local discrete energy combining unary foreground likelihood and pairwise smoothness/edge terms can find a lower-error boundary than thresholding or morphology.
Target population	Strict 1:1 lateral-deficit cases and healthy controls. A separate experiment may enlarge the ROI beyond endpoints if testing axial reach.
Inference-time inputs	YOLO/semantic foreground likelihood, raw image features, initial mask, local ROI.

Instrument validation

- Confirm ROI includes the missing truth in oracle analysis. If a narrow band excludes most missing pixels, graph cut cannot possibly recover them; report that as an ROI limitation, not a method failure.

Faithful implementation

23. Build binary labels y_i in a local ROI. Unary U_i should penalize disagreement with model/image evidence and optionally with the original mask.
24. Pairwise V_ij should penalize label discontinuities less across strong image edges and more within coherent filament regions.
25. Use s-t min-cut for submodular binary energies. Keep weights interpretable and small in number.
26. Run separate ROI widths, including a boundary band and an endpoint-expanded band.
27. Raster output replaces only the target instance; then rerun overlap resolution and official scorer.

Minimal viable experiment

One unary from semantic probability + one contrast-sensitive Potts pairwise term. Compare narrow boundary band and endpoint-expanded ROI.

Controls / ablations

- Unary only (thresholding)
- Pairwise only is not meaningful; instead compare constant pairwise versus contrast-sensitive
- Uniform morphology

Required metrics

- PQ delta
- Recovered missing pixels vs added false pixels
- Crossings
- Healthy-object harm
- Energy decrease versus PQ change (to detect surrogate mismatch)

Decision rule

Advance / accept: Keep if a fixed energy improves actual PQ on an inference-time target set and is robust to modest weight variation.

Reject / demote: Reject only the tested energy/ROI family after actual min-cut outputs are scored. A poor semantic unary does not rule out image-derived or ensemble unaries.

Common invalid conclusions to avoid

- Calling it tested because 59% of deficit is axial.
- Choosing ROI using truth.
- Assuming lower CRF energy implies higher PQ.

5. Vesselness / ridge filtering

Current evidence status: Global proposal use is weak; local endpoint evidence remains open and should be tested separately.

Hypothesis	Second-order image structure can reveal elongated ridge/valley continuation that the learned semantic field misses, especially near an existing endpoint.
Target population	Axial truncation endpoints plus healthy endpoints. Secondary test on complete misses is separate and should not be mixed with endpoint evidence.
Inference-time inputs	Raw/contrast channels at multiple scales; Hessian eigenvalues; optional dark-filament sign.

Instrument validation

- Validate that truth continuation passes through a local ridge/valley response more often than control directions. Use the best truth ray only for instrument validation, never inference.

Faithful implementation

28. Compute multiscale Hessian responses at scales matched to filament widths. Use Frangi/Sato-like response but choose bright-ridge vs dark-ridge sign correctly.
29. Sample response specifically inside the forward endpoint corridor or along candidate paths, not over the full solar disk.
30. Aggregate by max-over-scale, path mean, path minimum, or continuity length. Test both high-response and low-response directions because sign conventions can invert.
31. Optionally combine ridge orientation with endpoint tangent consistency.

Minimal viable experiment

Compute local multiscale ridge response for each endpoint, then ask whether the best orientation-consistent ray/path separates axial truncations from controls. No repair yet.

Controls / ablations

- Global ridge selectivity (already known weak)
- Random local directions
- Single-scale versus multiscale

Required metrics

- PR-AUC and precision at actionable coverage
- Conditional AUC among low-semantic-support endpoints
- Path continuity length
- Eventually PQ when used as a gate/cost

Decision rule

Advance / accept: Advance if local ridge evidence has useful precision or materially improves the path cost/gate over raw dark support alone.

Reject / demote: Reject local vesselness only after local, scale-appropriate, sign-correct measurements fail. Do not cite global 26.2% disk occupancy as a complete rejection.

Common invalid conclusions to avoid

- Testing only global threshold masks.
- Using wrong ridge polarity.
- Ignoring scale relative to filament width.
- Reporting ROC AUC without precision.

6. Orientation fields

Current evidence status: Untested directly and one of the highest-value remaining diagnostics.

Hypothesis	True filament continuation preserves local direction. Orientation agreement between endpoint tangent and nearby image structure can distinguish real continuation from arbitrary dark network.
Target population	Axial truncation endpoints versus healthy endpoints; optionally merge junctions for a second use case.
Inference-time inputs	Structure tensor or Hessian eigenvectors, centreline tangent, local coherence/anisotropy magnitude.

Instrument validation

- Truth continuation direction should align with estimated orientation more often in positive corridors than random/control corridors. Verify this oracle relation first.

Faithful implementation

32. Compute structure tensor J at 2-4 smoothing scales. Extract principal orientation and coherence c = (lambda1-lambda2)/(lambda1+lambda2+eps).
33. Because orientation is axial, compare modulo pi: agreement = |cos(2*(theta-theta_endpoint))| or equivalent.
34. Multiply directional agreement by coherence so isotropic/noisy pixels do not look informative.
35. Measure maximum continuous run of high agreement ahead of the endpoint and integrated orientation cost along candidate paths.
36. Use orientation as a cost term or gate; it need not be a standalone classifier.

Minimal viable experiment

For each endpoint, compute the best 4-30 px ray score from orientation agreement * coherence. Compare positives and healthy controls in both directions, then combine with dark_best_ray.

Controls / ablations

- Randomized endpoint tangent
- Rotated orientation field by 90 degrees
- Coherence removed

Required metrics

- AUC/PR-AUC
- Precision at fixed harm budget
- Incremental value over dark support via stratified tables or a simple fixed combination
- PQ after using it to gate continuation

Decision rule

Advance / accept: Advance if it improves precision or net PQ when added to raw support, even if its standalone AUC is modest.

Reject / demote: Reject only after multiscale orientation and coherence-aware scoring fail. A weak semantic/spine field does not test raw-image orientation.

Common invalid conclusions to avoid

- Using angle difference without pi-periodicity.
- Treating orientation in isotropic regions as reliable.
- Testing orientation alone and ignoring its intended role as a conditional cue.

7. Persistent homology / topological persistence

Current evidence status: Untested and lower priority; primarily a stabilizer for skeleton/branch decisions, not a solution to the dominant 1:1 shape error.

Hypothesis	True structural branches/connections persist across threshold or scale, while skeletonization artifacts disappear quickly. Persistence can prevent false split/merge repairs caused by tiny topological noise.
Target population	Masks with branch points, loops, or candidate splits; especially cases where skeleton topology changes under small perturbations.
Inference-time inputs	Probability field or distance transform across thresholds, skeleton/graph, filtration parameter.

Instrument validation

- Measure how known truth-supported branches persist as threshold/scale changes compared with spurious skeleton twigs.

Faithful implementation

37. Choose a filtration: semantic probability threshold, distance-transform level, or morphological scale.
38. Compute connected components/loops over filtration and persistence lifetimes (H0/H1 as relevant).
39. Associate persistent features back to spatial branches or connections.
40. Use persistence only to keep/remove candidate branches or validate a split; do not globally alter masks without a downstream operator.

Minimal viable experiment

On the merge/split subset, compare branch persistence of oracle-relevant versus irrelevant skeleton branches. This is a diagnostic before any PQ test.

Controls / ablations

- Branch length alone
- Single-threshold skeleton
- Random small perturbations

Required metrics

- Separation of relevant branches
- Stability under threshold perturbation
- False split reduction when inserted into a split operator
- PQ delta of the final operator

Decision rule

Advance / accept: Keep if persistence improves branch selection or reduces false repairs beyond simpler branch length/width statistics.

Reject / demote: Reject as unnecessary if it adds no value over simpler stability measures. Do not call it invalid because topology errors are a minority; that only lowers priority.

Common invalid conclusions to avoid

- Using persistence as a fancy scalar without mapping features back to actionable branches.
- Evaluating topological similarity but not downstream PQ.

8. Minimum-cost flow / global filament assignment

Current evidence status: Untested. Oracle headroom for fragmentation is modest, so this is a bounded secondary method.

Hypothesis	When multiple fragments/endpoints coexist, global assignment avoids greedy links that steal partners and can restore complete instances with consistent geometry.
Target population	N:1 fragmentation and N:M overlap-graph components. Do not mix with strict 1:1 shape failures.
Inference-time inputs	Fragment endpoints, distances, tangents, widths, path/image support, candidate link graph.

Instrument validation

- Using truth only, verify that the correct partner edge is present in the candidate graph. Report candidate recall before solving assignment. If the correct link is absent, assignment cannot succeed.

Faithful implementation

41. Build nodes for fragments/endpoints. Add an edge only when spatial distance, tangent compatibility, and ROI constraints are plausible.
42. Define cost C_ij = wd*d + wt*(1-cos delta_theta) + ww*|width_i-width_j| - ws*support(path). Normalize each term.
43. Solve minimum-weight matching or min-cost flow with constraints preventing one endpoint from being used twice and allowing unmatched endpoints at a penalty.
44. Union linked masks plus a connecting corridor only when the assignment chooses the edge.

Minimal viable experiment

Candidate graph + Hungarian/min-weight matching using distance+tangent only. Score actual unioned masks.

Controls / ablations

- Greedy nearest-neighbour linking
- Oracle partner link ceiling
- Distance-only assignment

Required metrics

- Candidate-edge recall
- Correct-link precision
- PQ delta
- Rescued fragmented truths
- False merges created

Decision rule

Advance / accept: Keep if global matching beats greedy linking and stays net-positive under the official scorer.

Reject / demote: Reject if even oracle-feasible components cannot be linked with a useful inference-time cost, or actual linking is net-negative. Small oracle ceiling may justify stopping for opportunity cost.

Common invalid conclusions to avoid

- Using per-truth maxima instead of global overlap structure.
- Evaluating link classification without rasterizing and scoring merged masks.

9. Matched filtering

Current evidence status: Untested.

Hypothesis	If filament cross-section/intensity is stereotyped, correlation with line/curve templates can detect weak continuation better than generic intensity thresholds.
Target population	Axial truncation endpoints and healthy endpoints; optionally complete misses as a separate discovery task.
Inference-time inputs	Raw/dark-contrast image, template bank over angles, widths, and lengths.

Instrument validation

- Estimate template width/intensity profiles from healthy predictions or training images without using target validation truth to tune per-instance templates.

Faithful implementation

45. Build zero-mean normalized templates representing dark elongated structures at several widths and orientations.
46. Compute normalized cross-correlation in a local endpoint ROI. Restrict orientations around endpoint tangent for continuation tests.
47. Use max response, response continuity, and length above threshold as features. Correct for multiple-template maxima by evaluating healthy controls identically.
48. Optionally curve templates gently for expected filament bending.

Minimal viable experiment

Straight dark-line templates, 5-7 angles around endpoint tangent, 2-3 widths. Audit local max response and continuous supported length.

Controls / ablations

- Random orientation templates
- Isotropic blob template
- Raw dark_best_ray

Required metrics

- PR-AUC
- Precision at low false-extension budget
- Incremental value over simple dark absorption
- PQ once used as a continuation gate

Decision rule

Advance / accept: Advance if matched response materially improves trigger precision or path selection.

Reject / demote: Reject the template family if responses cannot distinguish positives at actionable precision after width/orientation matching.

Common invalid conclusions to avoid

- Using unnormalized correlation so local brightness dominates.
- Searching the entire image and then comparing to a local endpoint task.
- Choosing template width from validation truth.

10. Anisotropic / coherence-enhancing diffusion

Current evidence status: Untested as preprocessing; should only be evaluated through downstream evidence quality.

Hypothesis	Smoothing preferentially along coherent filament direction can strengthen weak continuous structure while suppressing transverse noise, improving ridge/orientation/path cues.
Target population	Same endpoints used for ridge/orientation tests; not a separate prediction population.
Inference-time inputs	Raw image, structure tensor, diffusion parameters/time.

Instrument validation

- Verify that diffusion increases continuity along known healthy filaments without merging adjacent network structures.

Faithful implementation

49. Compute diffusion tensor from local structure tensor. Use stronger diffusion along the filament orientation and weaker across it.
50. Test a very small set of diffusion times. Preserve original image as a channel rather than replacing it outright.
51. Recompute the exact downstream cue (ridge response, orientation coherence, matched filter, geodesic cost) after diffusion.
52. Do not score diffusion by visual smoothness.

Minimal viable experiment

One coherence-enhancing diffusion setting followed by the orientation/ridge audit; compare identical metrics before/after.

Controls / ablations

- Gaussian smoothing with comparable noise reduction
- No diffusion

Required metrics

- Incremental trigger precision
- Path continuity
- False bridge rate between nearby structures
- Final PQ if adopted

Decision rule

Advance / accept: Keep only if it improves a downstream method and does not create false bridges.

Reject / demote: Reject preprocessing if downstream metrics are unchanged/worse across modest diffusion times. This says nothing about the downstream method itself.

Common invalid conclusions to avoid

- Calling diffusion a segmentation method.
- Evaluating PSNR/visual appearance instead of the final cue/PQ.
- Over-diffusing until separate structures merge.

11. Distance-transform / marker-controlled watershed

Current evidence status: Untested as proposed; directly relevant to the measured merge oracle but needs a reliable marker/gate.

Hypothesis	A merged prediction contains multiple interior width maxima/basins corresponding to separate filaments. Marker-controlled watershed on an appropriate energy can split it with fewer false cuts than skeleton neck heuristics.
Target population	1:N merge overlap-graph components, plus healthy 1:1 masks matched for area/shape as false-split controls.
Inference-time inputs	Predicted mask, Euclidean distance transform, optional image gradient/ridge map, markers.

Instrument validation

- Oracle test: do true constituent objects correspond to separable maxima/markers inside the merged mask? Measure marker recall and count before running watershed.

Faithful implementation

53. Compute distance transform D inside mask. Smooth lightly at scale below expected filament width.
54. Generate markers from persistent/local maxima, optionally constrained by skeleton endpoints or width-profile peaks.
55. Run marker-controlled watershed on -D, image gradient, or a combined energy. Restrict output to original mask so splitting does not change total foreground initially.
56. Post-merge tiny basins or basins below minimum support. Then score resulting instances globally.

Minimal viable experiment

Two-marker watershed on oracle-identified 1:N candidates, but markers must be generated by an inference-time rule. Compare -D watershed versus a simple neck cut.

Controls / ablations

- Uncontrolled watershed (expected over-splitting)
- Simple distance-transform maxima count
- Healthy masks

Required metrics

- Split precision
- Truths rescued
- Correct objects falsely split
- PQ delta
- Number of components produced per target

Decision rule

Advance / accept: Keep if a marker rule identifies a high-precision subset and watershed produces net-positive PQ.

Reject / demote: Reject a marker/watershed combination if false splits erase oracle gains. Do not reject watershed because a previous neck gate was weak.

Common invalid conclusions to avoid

- Letting truth provide markers in the reported deployed score.
- Scoring only split candidates and ignoring global false positives.
- Using every local maximum without persistence/min-distance suppression.

12. Medial-axis width-profile analysis

Current evidence status: Partially tested. Two scalar summaries are insufficient to rule out the richer profile.

Hypothesis	Merges create local, structured width anomalies (necks, saddles, multi-peak patterns) that global width CV and one minimum ratio can miss.
Target population	Oracle-splittable 1:N merges versus healthy 1:1 masks with comparable area/length; also compare unsplittable merges separately.
Inference-time inputs	Distance transform D, skeleton gamma(s), width series w(s)=2D(gamma(s)).

Instrument validation

- Ensure skeleton ordering is stable. Prune tiny branches or analyze main paths separately. Verify width samples correspond to physical cross-sections rather than skeleton artifacts.

Faithful implementation

57. Compute robustly smoothed width series along arc length.
58. Features: rho = min/median; saddle depth relative to adjacent peaks; prominence of minima; max |dw/ds| and mean |dw/ds|; number of prominent peaks; ratio of top two peaks; length fraction below q25; local width entropy; symmetry of expansion around a neck.
59. For branched skeletons, compute features per path between graph nodes and aggregate by max evidence rather than flattening all branches.
60. Evaluate each feature in both directions and a small interpretable rule such as deep saddle AND two strong adjacent peaks.

Minimal viable experiment

Run seven structure-preserving features plus the two existing controls. Then test one hand-specified two-feature gate; do not fit a high-capacity classifier first.

Controls / ablations

- Width CV
- rho alone
- Randomized width order (preserves histogram but destroys structure) - an excellent control for whether sequence order matters

Required metrics

- PR-AUC
- Precision at coverage
- Oracle-splittable merge recall
- False-split candidate rate on healthy masks
- PQ after a downstream split operator

Decision rule

Advance / accept: Advance if sequence-aware features improve precision over rho alone and identify a useful subset for watershed/neck cutting.

Reject / demote: Reject width-profile gating if even structure-aware statistics cannot reach useful precision. That still does not invalidate image-based split cues.

Common invalid conclusions to avoid

- Flattening a branched skeleton into arbitrary point order.
- Calling AUC 0.712 sufficient despite very low precision.
- Stopping at min/median or CV.

13. Shape derivatives / boundary sensitivity analysis

Current evidence status: Untested and research-oriented.

Hypothesis	Local boundary moves can be prioritized by the marginal change in a differentiable surrogate, allowing targeted expansion/shrinkage rather than global morphology.
Target population	Strict 1:1 shape errors; analyze lateral and axial separately because boundary parameterization differs.
Inference-time inputs	Boundary normal, semantic/image/ensemble energies, optional curvature.

Instrument validation

- The true PQ cliff is non-differentiable. Therefore first define a surrogate whose local derivative correlates with actual IoU/PQ changes on held-out perturbations. Validate that correlation before optimizing it.

Faithful implementation

61. Define energy E(M) = data fidelity + curvature + closeness to original mask. Derive or numerically estimate normal derivative dE/dn along boundary.
62. Perturb small boundary arcs outward/inward by 1-2 px and compare predicted energy change to actual validation IoU change (analysis only).
63. If the sensitivity map is calibrated, move only high-benefit arcs under an area/curvature budget.
64. Recompute official PQ after a fixed number of moves.

Minimal viable experiment

Finite-difference sensitivity rather than analytic shape calculus: test local arc perturbations and whether surrogate ranking predicts actual IoU improvement.

Controls / ablations

- Uniform outward move
- Random boundary arcs
- Semantic confidence alone

Required metrics

- Rank correlation between predicted sensitivity and actual IoU change
- PQ delta
- Crossings
- Healthy harm

Decision rule

Advance / accept: Proceed only if the surrogate sensitivity reliably ranks beneficial boundary moves and the greedy operator improves PQ.

Reject / demote: Reject the surrogate if local sensitivity has little relation to real metric changes. Do not conclude that all shape-derivative methods are impossible because PQ itself is discontinuous.

Common invalid conclusions to avoid

- Differentiating PQ directly as if it were smooth.
- Using truth-derived gradients in inference.
- Reporting energy decrease without metric gain.

14. Optimal transport between predictions

Current evidence status: Untested. Plain disagreement is only a cheap baseline, not OT.

Hypothesis	Two models may predict similar structures with systematic spatial displacement. Transport captures where foreground mass from one prediction would need to move to align with another, revealing directional boundary bias that XOR/disagreement loses.
Target population	Strict 1:1 masks where both architectures detect the same object; near-threshold positives versus healthy controls.
Inference-time inputs	Soft probability maps or normalized foreground mass from model A and B in a local ROI.

Instrument validation

- Establish reliable instance correspondence between the two models before computing transport. If correspondence is ambiguous, OT features are meaningless.

Faithful implementation

65. Crop a local ROI and normalize nonnegative probability mass. Use entropic Sinkhorn OT with ground cost equal to pixel distance, optionally anisotropic along filament orientation.
66. Extract total transport cost, net outward radial transport relative to model A boundary, axial transport along endpoint tangent, and lateral transport.
67. Compare against simple XOR and boundary disagreement. OT is justified only if it adds directional information.
68. If used for repair, transport vectors can suggest where to expand/shrink or which model to trust locally.

Minimal viable experiment

Compute Sinkhorn transport on matched instances for a small sample and test whether outward/axial transport separates near-threshold failures better than plain disagreement.

Controls / ablations

- XOR disagreement
- Difference of soft probabilities
- Centroid displacement only

Required metrics

- Incremental PR-AUC over disagreement
- Directional calibration
- PQ when used as a gate/refinement signal
- Runtime

Decision rule

Advance / accept: Keep only if OT contributes information unavailable from cheaper disagreement features.

Reject / demote: Reject as unnecessary if simple disagreement matches it. That is an efficiency decision, not evidence OT was conceptually wrong.

Common invalid conclusions to avoid

- Computing OT between unmatched instances.
- Normalizing away meaningful total-mass differences without also recording them.
- Claiming exp107 disagreement already tested OT.

15. Test-time energy minimization

Current evidence status: Untested by design; this is a late-stage framework that should integrate only components already shown useful.

Hypothesis	The final mask should reconcile several evidence sources while obeying geometric regularity, and joint optimization may outperform ad hoc sequential rules.
Target population	Only after useful evidence terms have been validated. Start on strict 1:1 shape errors before including split/merge topology.
Inference-time inputs	Original instance, model probabilities, raw image cues, orientation/ridge fields, ensemble predictions, curvature/shape priors.

Instrument validation

- Each energy term must have independent evidence that it correlates with a desired correction. If a term failed its standalone audit, exclude it rather than hiding it inside a large energy.

Faithful implementation

69. Define E(M) = sum_k lambda_k d_k(M,evidence_k) + lambda_curv R_curv(M) + lambda_orig d(M,M0).
70. Choose an optimization family consistent with topology: graph cut for binary local energies, level set for smooth boundaries, discrete proposals for topology changes.
71. Add terms one at a time. Maintain an ablation table. Keep lambdas on comparable normalized scales.
72. Use a fixed small iteration budget and deterministic initialization.

Minimal viable experiment

Combine exactly two proven cues plus a stay-close regularizer. Do not start with a six-term energy.

Controls / ablations

- Each term alone
- Original mask
- Sequential application of the same two cues

Required metrics

- PQ delta
- Ablation contribution of each term
- Stability across lambda perturbations
- Runtime
- Crossings/harm

Decision rule

Advance / accept: Adopt if joint optimization beats its strongest component and sequential baseline robustly.

Reject / demote: Reject the combined formulation if gains vanish under ablation or require brittle lambda tuning. Individual cues may still remain useful.

Common invalid conclusions to avoid

- Using energy minimization to obscure which signal helps.
- Tuning many lambdas on one validation split.
- Assuming lower energy means better PQ.

16. Combined endpoint geodesic + orientation support

Current evidence status: Not fairly tested. This is the most specific composite proposal and should be tested as a complete operator.

Hypothesis	A truncated endpoint can be extended safely when three conditions agree: image evidence supports a continuation, local orientation aligns with the endpoint tangent, and the integrated path remains geometrically smooth.
Target population	Strict 1:1 axial-truncation cases with healthy endpoint controls. This is narrower than all near-threshold masks.
Inference-time inputs	Endpoint/tangent, raw dark-support image, multiscale orientation/coherence, optional ridge evidence. Semantic probability should be optional because current measurements suggest it may be silent.

Instrument validation

- Truth-best-path or truth-best-ray must confirm that the forward search geometry contains the missing truth. Then separately verify that raw support and orientation each have at least weak conditional information.

Faithful implementation

73. Define state (x,y,heading). Transition to nearby pixels/headings with cost: a*(1-dark_support) + b*(1-orientation_agreement*coherence) + c*turn_penalty + d*length_penalty.
74. Use A* or dynamic programming in the endpoint ROI. Heading state avoids the failure mode where ordinary Dijkstra zig-zags among low-cost pixels.
75. Generate several candidate terminal lengths. Select using an inference-time stopping rule based on rising cost, loss of coherence, or path support - never truth.
76. Reconstruct width from the endpoint width profile; optionally taper slowly. Clip overlap with other confident instances or resolve competition explicitly.
77. Gate only when path quality exceeds a threshold chosen from a grouped validation sweep. Score the complete pipeline with official PQ.

Minimal viable experiment

Raw dark support + structure-tensor orientation + turning penalty. No semantic term. Three path lengths. Width fixed to local endpoint median. This directly tests whether the combination rescues axial truncations.

Controls / ablations

- Dark-only path
- Orientation-only path
- Straight tangent extension
- Semantic-only path
- Randomized orientation field

Required metrics

- PQ delta
- Axial near-threshold crossings
- Precision of trigger
- Healthy false extensions
- Path cost margin between selected path and second-best path
- Ablation gains

Decision rule

Advance / accept: Keep if the combined operator beats dark-only and straight-extension controls and remains net-positive under official PQ.

Reject / demote: Reject this composite only after the complete cost + path + mask reconstruction + gate is run and scored. Weak cone averages, zero semantic pixels, or weak orientation AUC alone are not equivalent tests.

Common invalid conclusions to avoid

- Calling exp109/110 a full test of the combination.
- Using a ray search when the proposed algorithm is allowed to curve.
- Forgetting heading state/turn cost.
- Scoring path evidence without actually adding pixels and rerunning PQ.

Appendix A - Standard experiment report schema

Every experiment should emit a machine-readable report.json and a short human-readable markdown entry. The report should make it impossible for a successor agent to confuse oracle labels, triggers, and deployed outputs.

```
{
  "experiment_id": "expXXX",
  "method": "geodesic_dark_orientation",
  "baseline_artifact_hash": "...",
  "population_definition": {
    "analysis_positive": "strict_1to1_iou_0.40_0.50_axial",
    "control": "strict_1to1_iou_ge_0.50_matched_size",
    "ground_truth_used_only_for_labels": true,
    "n_positive": 0,
    "n_control": 0
  },
  "instrument": {
    "label_feature": "truth_best_path_occupancy",
    "label_auc": null,
    "validated": false
  },
  "trigger": {
    "name": "path_cost_margin",
    "features": {},
    "both_directions_tested": true,
    "pr_auc": null,
    "operating_points": []
  },
  "operator": {
    "implemented": true,
    "description": "...",
    "uses_ground_truth": false,
    "parameters": {}
  },
  "official_metric": {
    "before": {"pq": 0.0, "tp": 0, "fp": 0, "fn": 0, "iou_sum": 0.0},
    "after":  {"pq": 0.0, "tp": 0, "fp": 0, "fn": 0, "iou_sum": 0.0},
    "delta_pq": 0.0
  },
  "harm": {
    "crossings_up": 0,
    "crossings_down": 0,
    "healthy_modified": 0,
    "healthy_below_0.5_after": 0
  },
  "conclusion": {
    "status": "partial",
    "rules_out": "...",
    "does_not_rule_out": "..."
  }
}
```

Naming rules that prevent self-pool mistakes

- Never call a feature "disagreement" when both inputs are the same pool. Name it self_consistency_* or same_pool_boundary_variation_*.
- Include model/pool identifiers in every feature key when two architectures are involved, e.g. disagreement_exp002_vs_exp097.
- Assert other_pool_id != pool_id when the experiment claims cross-architecture disagreement. If equality is intentional, require an explicit --allow-self-pool flag.
- Write source checkpoint hashes and preprocessing signatures into report.json so a successor can reproduce the exact field.

Reproducibility assertions

```
assert baseline_pq == expected_baseline_within_tolerance
assert all_prediction_ids_unique
assert all_truth_ids_unique
assert population_counts_recomputed_from_source
assert not (experiment_claims_cross_architecture and pool_id == other_pool_id)
```

```
assert not deployed_operator_reads_ground_truth
assert official_scorer_ran_after_operator
assert report_contains_before_and_after_tp_fp_fn
```

Appendix B - Decision tree and stopping rules

Step 1: Is there oracle headroom? - If no, stop that repair family. If yes, quantify the maximum PQ ceiling and continue.

Step 2: Is the target population structurally identifiable? - Test inference-time cues with both directions and matched controls. If no cue reaches useful precision, either search a genuinely different evidence source or stop trigger-based post-processing.

Step 3: Does the cue test the actual algorithmic condition? - If not, redesign the audit. Path methods need path statistics; sequence methods need sequence-aware statistics; global optimization needs actual assignments.

Step 4: Does a faithful minimal operator improve local outcomes? - Rasterize the repair. Count crossings and harm. If the local operator is clearly destructive, stop before a large sweep.

Step 5: Does official PQ improve? - If no, the method is not useful in its current form even if local AUC/Dice improves.

Step 6: Is the gain stable? - Check images, modest parameter perturbations, and preferably grouped resampling. Brittle one-point gains should be treated cautiously.

Step 7: Only then compose methods. - A combined energy/pipeline should contain components with individually demonstrated value, unless the interaction itself is the hypothesis and is tested explicitly.

When to stop searching for a trigger

If several genuinely independent evidence sources all fail at actionable precision and the actual repair operator has no net-positive region, stop trying to identify the same failure with more scalar features. Redirect effort to (a) a structurally identifiable smaller error class, (b) base-model changes, or (c) a different representation. The stopping criterion should be evidence convergence, not frustration.

When a negative result is strong enough to call a method invalid for this project

- The tested implementation is faithful to the core decision rule of the method.
- The inputs and preprocessing match the intended signals exactly.
- The target population contains the failure type the method is designed to address.
- The instrument was validated with a truth-only label where possible.
- Both score directions and reasonable scales/parameters were tested.
- The actual operator was applied and the official scorer rerun.
- No positive PQ region exists without unacceptable collateral damage.
- The report states the scope: "this formulation on this dataset," not a universal mathematical claim.

Appendix C - Checklist before declaring a method invalid

- [ ] **01.** Did I implement the actual method, or only a proxy?
- [ ] **02.** If a proxy, is failure of that proxy logically necessary for the method to work?
- [ ] **03.** Did I validate the geometry/instrument with a truth-only label?
- [ ] **04.** Did I test both high and low feature directions?
- [ ] **05.** If AUC < 0.5, did I inspect the flipped direction?
- [ ] **06.** Did I report precision-recall, not just ROC AUC?
- [ ] **07.** Did I match control objects by relevant morphology/image factors?
- [ ] **08.** Did I recompute population counts from source artifacts?
- [ ] **09.** Did I check for same-pool/self-consistency mislabeled as cross-model disagreement?
- [ ] **10.** Did I reproduce the exact preprocessing/checkpoint that generated any saved field?
- [ ] **11.** Did I run the actual repair operator?
- [ ] **12.** Did I rasterize the full mask, not only score a centreline/path?
- [ ] **13.** Did I rerun global matching and official PQ?
- [ ] **14.** Did I count healthy objects harmed and threshold crossings down?
- [ ] **15.** Did I distinguish oracle ceiling from inference trigger?
- [ ] **16.** Did I avoid using truth to select parameters/stopping distance per instance?
- [ ] **17.** Did I compare against a simpler control with equal area/complexity?
- [ ] **18.** Did I test a reasonable, interpretable parameter range?
- [ ] **19.** Did I record what remains untested?
- [ ] **20.** Would a successor reading report.json understand exactly what was and was not ruled out?

Final instruction to the experimental agent: Negative results are valuable only when they kill the right hypothesis. Prefer a slower, faithful test over a fast proxy that answers a different question. The goal is not to accumulate rejections; it is to reduce uncertainty about what can improve PQ.